

OSSP SKILL-Q4

- To Split Input into Tokens, Identify Delimiters, Handle Whitespace, Create Token Structures, Validate Token Streams, Debug Parsing Output.

To Design Parser Logic, Generate Parse Trees, Validate Syntax, Detect Errors, Handle Empty Commands, Produce Execution Structures.

Code:-

```
#include <stdio.h>

#include <string.h>

#include <stdlib.h>

#include <ctype.h>

#define MAX_INPUT 1024

#define MAX_TOKENS 100

int main()

{

    char input[MAX_INPUT];

    char tokens[MAX_TOKENS][MAX_INPUT];

    int tokenCount = 0;

    printf("===== Mini Shell Quote Parser =====\n");

    printf("Type 'exit' to quit.\n\n");

    while (1)

    {

        printf("MiniShell> ");

        if (fgets(input, sizeof(input), stdin) == NULL)

            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)

            break;
```

```
if (strlen(input) == 0)
    continue;
tokenCount = 0;
int i = 0;
while (input[i] != '\0')
{
    while (isspace(input[i]))
        i++;
    if (input[i] == '\0')
        break;
    int j = 0;
    if (input[i] == '\n')
    {
        i++;
        while (input[i] != '\n' && input[i] != '\0')
        {
            tokens[tokenCount][j++] = input[i++];
        }
        if (input[i] == '\n')
            i++;
    }
    else if (input[i] == '\"')
    {
        i++;
        while (input[i] != '\"' && input[i] != '\0')
        {
```

```
if (input[i] == '$')
{
    i++;
    char var[100];
    int k = 0;
    while (isalnum(input[i]) || input[i] == '_')
    {
        var[k++] = input[i++];
    }
    var[k] = '\0';
    char *value = getenv(var);
    if (value != NULL)
    {
        strcpy(&tokens[tokenCount][j], value);
        j += strlen(value);
    }
}
else
{
    tokens[tokenCount][j++] = input[i++];
}
}
if (input[i] == '"')
    i++;
}
else
```

```

{
    while (input[i] != '\0' && !isspace(input[i]))
    {
        tokens[tokenCount][j++] = input[i++];
    }
}
tokens[tokenCount][j] = '\0';
tokenCount++;
}
printf("\nParsed Tokens:\n");

for (int x = 0; x < tokenCount; x++)
{
    printf("Token %d : %s\n", x + 1, tokens[x]);
}

printf("-----\n");
}
printf("Shell Closed.\n");
return 0;
}

```

OUTPUT:-

```
#include <stdio.h>
#include <string.h>
#include <stdlib.h>
#include <ctype.h>

#define MAX_INPUT 1024
#define MAX_TOKENS 100

int main()
{
    char input[MAX_INPUT];
    char tokens[MAX_TOKENS][MAX_INPUT];
    int tokenCount = 0;

    printf("==== Mini Shell Quote Parser ==== \n");
    printf("Type 'exit' to quit. \n \n");

    while (1)
    {
```

^G Get Help	^O WriteOut	^R Read File	^Y Prev Pg	^K Cut Text	^C Cur Pos
^X Exit	^J Justify	^W Where is	^V Next Pg	^U UnCut Text	^T To Spell

```
apple@apples-MacBook-Pro-2 skill % ./skillq4
```

```
==== Mini Shell Quote Parser ====
Type 'exit' to quit.
```

```
MiniShell> ls
```

```
Parsed Tokens:
```

```
Token 1 : ls
```

```
-----
MiniShell> ls -l
```

```
Parsed Tokens:
```

```
Token 1 : ls
```

```
Token 2 : -l
```

```
-----
MiniShell> exit
```

```
Shell Closed.
```